

CHARCUBE · DESIGN DECISIONS REGISTER

CharCube design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-09-30	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

CharCube design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md` , CCB-BLD-001) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Design for construction: accept the twelve changes that make the concept buildable (pinwheel plinth, lid hole and collar, riveted shroud and 60 mm band, port damper guides, brick positions, retort handles, fins and bolted angle legs in place of the ring seat, 650 mm flue on stop clips with a riveted rain cap, baffle hanger rod, located jacket lid and vent, logger beside the leg, probes in through the side)	Accept; amend any change	Accept: none changes what the kiln does, its pitch or its safety case	The whole build plan	CCB-DDR-003, Table 1
2	Concrete blocks under a hot drum floor may crack or spall	(a) build as modelled and measure the block temperature on the first burn; (b) a 25 mm ceramic fibre board between pan and blocks	(a), then (b) if the blocks pass about 150 °C	Plinth (build plan section 3.1, first checks)	CCB-DDR-003, A2
3	Guiding the jacket sleeve onto the throat (2 mm clearance, 47 kg unit lowered by two people)	(a) as modelled; (b) flare the bottom 20 mm of the sleeve to about 190 mm	(b), a small extra for the welder	Water jacket (section 3.11), step 15	CCB-DDR-003, A3
4	Is the 47 kg two-person lift of the heat-recovery unit acceptable to users, or is a swinging arm needed?	(a) two-person lift as modelled; (b) a swinging arm on a post	None yet; ask users through a co-design partner	Tripod and lifts (sections 3.15, 3.16)	CCB-PRC-001, open questions
5	First region, residue and partner for co-design	Region and residue of the first adopting partner	None (portfolio rule: partners are chosen per area later)	Feedstock used for the first batches; named suppliers	CCB-DDR-001 item 9; CCB-DDR-002 Table 2
6	Rain cap shape: the photoreal renders show a conical cap; the model and drawings a flat 260 mm square	(a) adopt the cone in the model and drawings; (b) keep the flat cap	(a): a cone sheds rain better and the BOM does not fix the shape	Rain cap (section 3.18)	REVIEW 2026-09-26, appearance item 1
7	Hot-surface labels shown on the renders are not in the BOM	(a) add them to BOM line 14; (b) leave them out	(a): they cost little and support the safety section	Bought fixings (section 3.21)	REVIEW 2026-09-26, appearance item 5

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The two drums' actual diameters and heights, and the gap between them (at least 40 mm)	Drum sizes vary by maker; the gas gap, brick positions and probe slot height depend on them	CCB-DDR-003
2	The 160 mm pipe's actual outside diameter and the 168 mm sleeve's bore	The collar is rolled to the pipe; the slip joints need about 2 mm clearance	CCB-DDR-003, P2 and P8
3	The effect of the 150 mm lid hole on draft (8 % less area than the throat bore)	The draft calculation does not model the lid entry; margin 2.4 at peak	CCB-DDR-003, P2; CCB-CAL-001 section 6
4	Probe lengths (500 mm and 300 mm) and sheath rating	The core probe must reach the retort axis from outside the blanket	CCB-DDR-003, P12
5	Blanket roll widths (681 mm drum strip, 540 mm jacket strip)	Sets how many joins each wrap has	Build plan sections 3.4, 3.14
6	Prices of every BOM line once quoted, including the parts added for construction	The estimated cost against the value-engineering target rests on indicative prices	CCB-CAL-001 section 11
7	The factor of 3 on gas-side heat transfer and the four velocity heads of pressure loss assumed for the spiral baffle	Heat to water (R6) and draft depend on them; they are assumptions until measured at TRL 4	CCB-CAL-001 sections 1, 7

■ Value engineering

Value-engineering target: USD 320 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 319 as priced (USD 1 under the target); the parts added for construction (U-bolts, fins, foot cleats and pads, guide strips, clips, rod, bolts) are not yet priced and probably add USD 5 to 10, which would put the estimate USD 4 to 9 over the target.

- **Main cost drivers:** the ceramic fibre blankets, the welded water jacket with its spiral baffle insert, the two drums, and the concrete plinth blocks and firebricks (see `bom/bom.csv`).
- **Savings worth trying:** a cheaper logger box; a clay and ash render in place of the side blanket (about USD 25, from the TRL 2 review); and re-pricing the added small parts at purchase.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items 1 to 8 and 10: open-vented annular water jacket; \$300 kiln budget with the safety kit separate; nested-drum retort; bundled straw and stalks first; burner throat with preheated air; logger as standard; ceramic fibre blanket; tripod-carried jacket; keep the name	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	CCB-DDR-001
2026-09-25	Items 11 to 16: air shroud in place of the 25 mm ring; spiral baffle insert and jacket blanket; lid and top band blanket; R5 relaxed to 5 h; R9 restated to 700 °C; safety kit stated in R10 and the BOM notes	Amish: "i accept all your recommendations, go with them across all repos."	CCB-DDR-002
2026-09-26	Item 17: budget top-up from \$300 to \$320 for kiln parts	Amish: "I am ok with the budget top ups."	CCB-DDR-002, Table 1a
2026-09-25	TRL 4 on hold	Amish	<code>project.yaml</code> ; CCB-DDR-001
2026-09-30	Make the design physically buildable while drawing the build plan; outstanding decisions kept in this register, not in the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	CCB-DDR-003 (its changes are Draft, open decision 1)